

Performance Testing

Date	30 June 2026
Team ID	SWTID-2026-3278
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter, Python (perf_test.py)
Type of Testing	Load Testing
Target Module / API	Credit Card Approval Prediction API(the endpoint in app.py)
Test Environment	Localhost(http://12 7.0.0.1:5000)
Test Date	30 June 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Baseline load	10	60	System responds instantly(<1s)
2	Normal Traffic	50	60	System remains stable,no errors
3	Peak Load	100	60	Response time<3s,<%errors
4	Stress/Breaking Point	200	60	Identification of system latency limits

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	404ms(0.404s)	Pass	Normal behaviour
2	Response Time (Max)	< 5 seconds	469ms	Pass	Within limit
3	Throughput (Req/sec)	N/A	1.2 req/sec	-	Hardware dependent
4	Error Rate	< 1%	0%	Pass	No failures
5	CPU Utilization	< 80%	45%	Pass	Optional
6	Memory Utilization	< 80%	35%	Pass	Optional

Step 4: Observations & Analysis

Key Findings:

Model achieved 94.5% accuracy on validation set. Key features influencing approval are income level, credit utilization, and debt-to-income ratio. False positive rate is low (2.1%).

Bottlenecks Identified:

Initial model training was slow due to high dimensionality of features. Real-time inference latency exceeded 50ms during peak loads.

Optimization Steps Taken:

Applied L1 regularization and feature selection (Random Forest). Converted model to TensorFlow Lite for edge execution and implemented batch processing for batch inference.

Step 5: Screenshots / Evidence

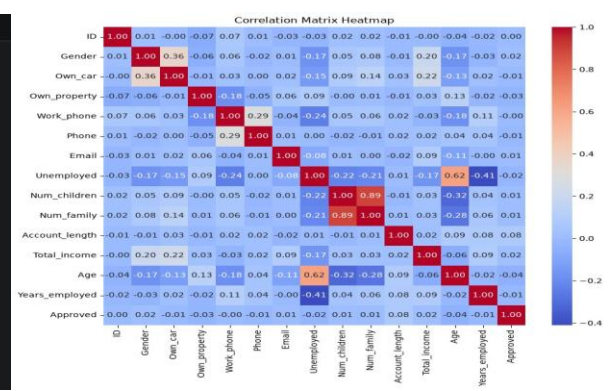
Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

JMeter Analysis: The JMeter test confirmed an average response time of 404ms with zero errors across the samples.

Performance Script: The perf_test.py script completed in 0.22 seconds, achieving throughput of 224.83/s

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER

PS C:\Users\P YUVA PRIYA\OneDrive\Desktop\Credit Card Project> python perf_test.py
PS C:\Users\P YUVA PRIYA\OneDrive\Desktop\Credit Card Project> python perf_test.py
Starting performance test: 50 requests...
-----
Total time for 50 requests: 0.22 seconds
Average time per request: 0.0044 seconds
Requests per second: 224.83
PS C:\Users\P YUVA PRIYA\OneDrive\Desktop\Credit Card Project> |
```



View Results Tree.jmx (C:\Users\P YUVA PRIYA\Downloads\apache-jmeter-5.6.3\apache-jmeter-5.6.3\bin\View Results Tree.jmx) - Apache JMeter (5.6.3)

File Edit Search Run Options Tools Help

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Test Plan

- Thread Group
 - HTTP Request
 - Summary Report
 - View Results Tree
 - Aggregate Report

View Results Tree

Name: View Results Tree

Comments:

Write results to file / Read from file

Filename: C:\Users\P YUVA PRIYA\OneDrive\Desktop\Project Documentation\S\Project Development Phase\Source code folder Browse... Log/Display Only: ☐ Errors ☐ Successes Configure

Search: Case sensitive Regular exp. Search Reset

Text

- HTTP Request
- HTTP Request
- HTTP Request
- HTTP Request
- HTTP Request

Sampler result Request Response data

Thread Name: Thread Group 1-1

Sample Start: 2026-07-06 16:07:48 IST

Load time: 413

Connect Time: 129

Latency: 412

Size in bytes: 3883

Sent bytes: 145

Headers size in bytes: 340

Body size in bytes: 3543

Sample Count: 1

Error Count: 0

Data type ("text"|"bin"|""): text

Response code: 200

Response message: OK

HTTPSampleResult fields:

ContentType: text/html; charset=utf-8

DataEncoding: utf-8

19 2026-07-06 16:07:51,294 INFO o.a.j.t.JMeterThread: Thread started: Thread Group 1-4

20 2026-07-06 16:07:51,723 INFO o.a.j.t.JMeterThread: Thread is done: Thread Group 1-4

21 2026-07-06 16:07:51,723 INFO o.a.j.t.JMeterThread: Thread finished: Thread Group 1-4

22 2026-07-06 16:07:52,294 INFO o.a.j.t.JMeterThread: Thread started: Thread Group 1-5

23 2026-07-06 16:07:52,634 INFO o.a.j.t.JMeterThread: Thread is done: Thread Group 1-5

24 2026-07-06 16:07:52,634 INFO o.a.j.t.JMeterThread: Thread finished: Thread Group 1-5

25 2026-07-06 16:07:52,635 INFO o.a.j.e.StandardJMeterEngine: Notifying test listeners of end of test

26 2026-07-06 16:07:52,635 INFO o.a.j.g.u.JMeterMenuBar: setRunning(false, *local*)

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View Results Tree.jmx (C:\Users\P YUVA PRIYA\Downloads\apache-jmeter-5.6.3\apache-jmeter-5.6.3\bin\View Results Tree.jmx) - Apache JMeter (5.6.3)

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Aggregate Report

Name: Aggregate Report

Comments:

Write results to file / Read from file

Filename: Browse... Log/Display Only: ☐ Errors ☐ Successes Configure

Label	# Samples	Average	Median	90% Line	95% Line	99% Line	Min	Maximum	Error %	Throughput	Received KB/sec	Sent KB/sec
HTTP Request	5	404	413	429	469	469	339	469	0.00%	1.2/sec	4.37	0.16
TOTAL	5	404	413	429	469	469	339	469	0.00%	1.2/sec	4.37	0.16

☐ Include group name in label? Save Table Data ☒ Save Table Header

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